

BREATHE-CC · INTEGRATED EXPOSURE INDEX

More exposure detail, without more burden

Proposed additions to the monthly follow-up, and what they cost a family

Aidan Meyers · Rollins College Honors Thesis · Melaram Lab, TAMU-CC
September 28, 2026

What is being added

All of it inside the existing monthly follow-up. No existing question changes its wording, choices or answer options.

- | | | |
|----------|--|---|
| A | Daily routine | Where the child is through a typical weekday and weekend, in seven time blocks, plus time in a car. |
| B | Time between homes | For two-home families: which schedule they follow, and two anchors that let us project nights at each home. |
| C | Home habits, smoking and vaping | Cooking frequency at each home; smoking carried forward from baseline; vaping, which we have never asked. |
| D | This past month | Windows open, nights at each home this month, break stays, storms and outages. |

Plus one change to an existing section: the medication questions would show last month's list and ask whether anything changed. That one needs Mari's call (slide 10).

What it costs a family, in taps

Counted by running the data dictionary’s own branching logic. A tap is one answer; the routine grid counts as seven.

7

taps in an ongoing month, one home
Nothing changed, just confirmations

10–11

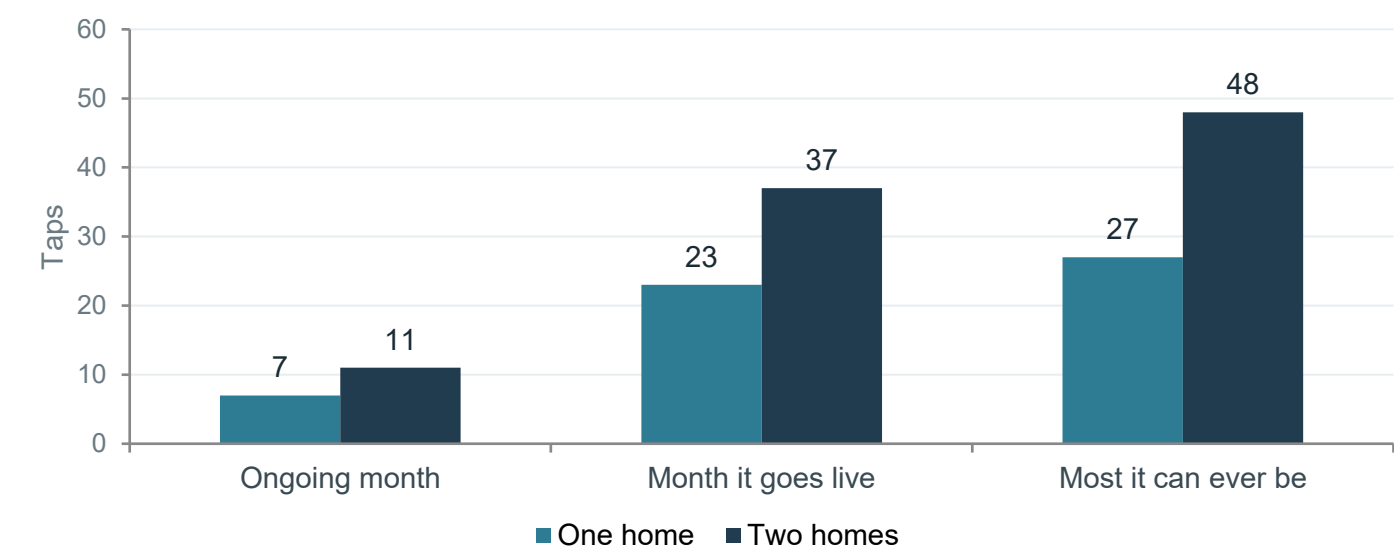
taps in an ongoing month, two homes
Adds the schedule check

23

taps the month it goes live, one home
14 of them are the two routine grids

33–37

taps the month it goes live, two homes
Once, then it carries forward



Against today’s form

- The monthly survey has 38 answerable fields today, 19 of them always shown.
- If the medication review is approved it removes 3 to 10 questions in a no-change month, so an ongoing month lands between -3 and +4 against today.
- Families are told before the longer month, by text and email, in both languages.

Two-home ranges span the six schedule patterns. Maximum assumes a break stay, smoking and vaping yes, and a new start date.

Why it does not compound

Everything the family tells us is stored per family and shown back the next month.

1

Asked once

The month the change goes live, the family fills in the routine, the schedule and the home questions.

2

Saved per family

A hidden instrument keeps the latest answer for each item, with the date it was given.

3

Confirmed after

Later months show a summary and ask: is this still true? One tap and they are done.

4

Missed months are safe

A skipped survey changes nothing. The next one shows the last saved answers, however old.

What a parent actually sees in month two

"Your child's routine on file (updated 08/20/2026). Weekday: ● Home, ● Home, ■ School ... Time in a car or bus: about 30 minutes a day."

→ Is this still your child's usual routine? ○ Yes, still the same ○ Something changed, let me update it

Section A · The child's day

A

Seven time blocks, weekday and weekend, plus time in a vehicle.

What changed

- Four columns: home inside, school or childcare inside, outside, and part inside part outside.
- The organized-sports column is gone; sports counts as outside.
- Symbols now carry their word (● Home, ■ School), so no one reads a key.
- A start date lets a routine be dated back to the start of the school year, never before enrollment.

Why we ask it

- Hourly monitor data is useless without knowing where the child was.
- Published precedent: multi-site, time-weighted exposure shifts effect estimates against a home-only assignment.
- School and break schedules are published, so we do not have to ask for what a calendar can tell us.

What it buys the index

- Time-activity weights for every hour of the month.
- Indoor versus outdoor splits, which decide how much the outdoor surface counts at all.
- A routine that carries month to month instead of being re-entered.

Section B · Time between homes

B

Two-home families only. Six schedule patterns, each with a projection rule.

What changed

- Six options drawn from Texas statute and court-published model schedules, including the standard and expanded possession order and the 2-2-3 and 2-2-5-5 rotations.
- Two anchors: where the child sleeps tonight, and how many nights until the next switch.
- A monthly one-question check against the projected schedule; a mismatch goes to a staff text.

Why we ask it

- Percent-of-time at each home is not how custody actually works.
- Both homes are geocoded, so nights at each home change the outdoor exposure assignment directly.
- Asking for every night each month would be unusable; asking the pattern once is not.

What it buys the index

- A projected calendar of nights at each home, checked monthly.
- Handoff days count as half a day at each home, so no handoff-time question is needed.
- Break stays are captured as a date range instead of a nightly diary.

Section C · Home habits, smoking and vaping

C

Both homes, with each address piped in so the family knows which one they are answering about.

What changed

- Cooking frequency stays; the air purifier moves to the 3-month safety form; candles and car windows are cut.
- Smoking is carried forward from baseline and confirmed monthly.
- Vaping is asked for the first time: whether anyone vapes inside, on how many days a week, and how many people.

Why we ask it

- Baseline asked about smoking once, up to six months ago, and never asked about vaping at all.
- Secondhand aerosol is a real but smaller indoor source than cigarettes, so it needs its own items rather than being folded in.
- Published cohort work indexes vaping on days per week and number of household users, which is what we ask.

What it buys the index

- An indoor source term that can change month to month, instead of a single baseline value carried for two years.
- Cooking frequency turns a yes-or-no gas stove into something that can be weighted.
- A vaping variable no one in this cohort has.

Section D · This past month

D

The only genuinely month-to-month items.

What changed

- Windows open at each home stays; how often the AC runs is cut.
- One question on where the child sleeps tonight, as a check on the projected schedule.
- Storms, outages and generator use stay, at Dr. Melaram's request.
- The school or childcare question moves to the 12-month survey only.

Why we ask it

- Infiltration is what decides how much of the outdoor concentration reaches the child indoors.
- HVAC type is already on file and predicts use; a parent with a smart thermostat cannot answer how often it runs.
- Outage data cannot be pulled by address here: the Texas grid has many retail providers.

What it buys the index

- A monthly infiltration term that follows the weather.
- A QC check that catches a schedule that has quietly changed.
- Short, sharp events that can explain an exacerbation that the monthly average cannot.

What we took out on September 28

Every cut below came out of the review with Dr. Warden, Dr. Melaram and Dr. Jin.

Air purifier

Cannot be weighted without make, size and use. Moved to the 3-month safety form as a mitigation question.

Candles and incense

No way to quantify candle exposure, and the incense literature comes from settings unlike ours.

Car windows open

Almost no coverage expected in South Texas; not enough to model against the burden.

How often the AC runs

HVAC type at baseline already predicts use, and smart thermostats make it unanswerable.

Organized sports column

Folded into outside, which is what it measures for exposure.

School change, monthly

Moved to the 12-month survey, where a published calendar can date it.

Six items cut, and the two routine grids trimmed from five columns to four — before this deck reached Driscoll.

The one piece that needs Mari's call

Everything else adds new questions. This one changes how an existing section is presented.

The proposal: the monthly form shows the medicines reported last time and asks "is this still what your child is taking?" Yes skips the medication questions for that month; No shows them exactly as they are today. Nothing is reworded.

Dr. Warden's objections

- It changes how an existing section is presented mid-study, which is a harder IRB argument than adding questions.
- Medication reporting may look more complete after the switch than before, creating a sub-cohort effect.

What we would do about it

- Medication is EHR-verifiable every month, which is why this section was chosen and not another.
- Flag the switch month in every record and report a before-and-after sensitivity analysis on reporting completeness.
- Dropping it later costs two fields and one line of branching.

If it stays, it removes 3 to 10 medication questions in a month where nothing changed — the single biggest burden reduction in the proposal.

Recommendation: put it to Mari with both objections stated, and drop it without argument if she is uneasy.

What this does to the data

The methodology questions raised in the review, and the answer to each.

Does it change the study mid-stream?

No existing question changes wording, choices or options. New questions are added, and the month they go live is flagged in every record.

Six months of baseline smoking

The baseline answer is piped in and confirmed rather than re-asked. Going to Driscoll: is that acceptable, or should smoking be prospective only from launch?

Back-dating routines

Dates come from published school calendars where possible, never before enrollment, and every back-dated value carries a flag so it can be dropped.

More variables than the sample supports (Dr. Jin)

Items feed two sub-indices, not one model term each. Structural blanks are not-applicable, not missing. Where the count is still high, items collapse into constructs.

Validation with this sample size

The thesis compares the index against the nearest-monitor approach on this cohort and states the limit. Fuller validation, including simulated data, belongs to the methods paper.

A swap on the 3-month form, at no net cost

The safety perception form already repeats three questions the monthly form asks every month.

Asked twice today

- Visible mold, in the past 3 months
- Damp spots, in the past 3 months
- Water damage, in the past 3 months
- All three are already on the monthly form, every month.



What we would ask instead

- What the household actually uses: purifier, allergen-proof mattress or pillow covers, HEPA vacuum, pest control, no smoking indoors
- How often a child was kept inside because of bad air
- Where the family hears about air quality
- What would help, including a bad-air-day alert or dashboard
- Whether a morning forecast would change their decisions

Answers the question reviewers keep asking — what do families do about it — and tells us whether an alert would change anything. Hiding the three duplicates keeps every answer already collected.

What we need, and when

Two decisions block the IRB submission. Everything else is ready to move.

- 1 Mari** The medication review: in or out.
- 2 Driscoll** Smoking: confirm the six-month-old baseline answer, or start fresh at launch.
- 3 The team** Hide the three duplicated indoor-air questions on the 3-month form and add the mitigation questions.

Then, in order

Copy of the project with the records, loaded and tested → printed surveys and data dictionary for the IRB packet → new question text to translation, with the participant notice → launch at the next monthly invitation.

The whole proposal in one line

**Ask once, confirm after: 7 taps a month for a one-home family,
and an exposure estimate that follows the child instead of the address.**

Full question-by-question document, demo data dictionary and a clickable preview of the survey are with this deck.

BREATHE-CC@tamucc.edu · www.melaramlab.com